

# Practical 2 - Non-parametric analysis of survival data

## Survival Analysis

London School of Hygiene & Tropical Medicine

## SOLUTIONS

Datasets required: `pbcbase`, `whitehall`

R packages required: `survival`, `ggplot2`, `survminer`

```
library(survival)
library(ggplot2)
library(survminer)
```

## Introduction

This practical is in two parts. We will use the Primary Biliary Cirrhosis (PBC) data and the Whitehall Study data, which are familiar to you from Practical 1.

Part A. You will use the Primary Biliary Cirrhosis (PBC) data and estimate survival probabilities and compare survival curves between treatment groups.

Part B. You will use the Whitehall data to investigate the effect changing the origin and entry dates has on survival curves.

## Aims

The aims of this session are:

- Calculate and interpret estimates of survival probabilities from data with censoring, using the Kaplan-Meier estimator.

- Construct and interpret Kaplan-Meier plots.
- Perform and interpret a log rank test.
- Understand the impact of changing the analysis timescale on survival estimates.

## Part A: Primary Biliary Cirrhosis data

The first dataset we will use is called `pbcbase`. This is data from a multicentre, double-blind, clinical trial for the treatment of Primary Biliary Cirrhosis (PBC). There are 184 individuals in the data and 17 variables; the variables of interest in this session are described below.

| Variable             | Description                                                         |
|----------------------|---------------------------------------------------------------------|
| <code>id</code>      | Unique identifier for each participant                              |
| <code>datein</code>  | Date person entered the study                                       |
| <code>dateout</code> | Date of the end of follow-up due to either death or censoring       |
| <code>d</code>       | Event indicator at the end of follow-up: 0=alive (censored), 1=dead |
| <code>time</code>    | Follow-up time in years                                             |
| <code>treat</code>   | Treatment: 1=placebo, 2=active                                      |

Load the data using:

```
pbcbase<-read.table("datasets/pbcbase.csv",sep="," ,header=T)
```

### Question 1

We will begin by obtaining Kaplan-Meier estimates of overall survival. In R you can create Kaplan-Meier estimates of survival probabilities  $S(t) = \Pr(T > t)$  using:

```
pbcbase.km <- survfit(Surv(time,d)~1,data=pbcbase)
```

An alternative is to use:

```
pbcbase.km <- survfit(Surv(as.numeric(dateout),d,
                           origin=as.numeric(datein))~1,data=pbcbase)
```

- You can see the Kaplan-Meier estimates of survival probabilities using `summary(pbcbase.km)`. What is the estimated probability of survival beyond 1 year? 5 years?
- Using the Kaplan-Meier estimator for  $S(t)$ , complete the table below to obtain estimates of  $S(t)$  for the first 5 event or censoring times in the data. Check your answers against what is shown in the R output.

| Event and censoring<br>times $t_j$ | Number at<br>risk | Number of<br>events | Number of<br>censorings | $\hat{S}(t_j)$ |
|------------------------------------|-------------------|---------------------|-------------------------|----------------|
| 0.0219                             | 184               | 0                   | 1                       |                |
| 0.0246                             | 183               | 2                   | 0                       |                |
| 0.0520                             | 181               | 1                   | 0                       |                |
| 0.1040                             | 180               | 2                   | 0                       |                |
| 0.1314                             | 178               | 1                   | 0                       |                |

## SOLUTION

```
#Kaplan-Meier analysis
pbc.km <- survfit(Surv(time,d)~1,data=pbc)

#Look at survival probability estimates
#Not shown in the interests of space
# summary(pbc.km,censored = T)

#we can obtain the estimates at specific times using
summary(pbc.km,times=c(1,5))
```

Call: survfit(formula = Surv(time, d) ~ 1, data = pbc)

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 1    | 159    | 18      | 0.901    | 0.0221  | 0.859        | 0.946        |
| 5    | 67     | 55      | 0.541    | 0.0407  | 0.467        | 0.627        |

- The estimated probability of survival beyond one year is 90.1% (95% CI 85.9 to 94.6). The estimated probability of survival beyond 5 years is 54.1% (95% CI 46.7 to 62.7).

| Event and censoring<br>times $t_j$ | Number at<br>risk | Number of<br>events | Number of<br>censorings | $\hat{S}(t_j)$                     |
|------------------------------------|-------------------|---------------------|-------------------------|------------------------------------|
| 0.0219                             | 184               | 0                   | 1                       | 1                                  |
| 0.0246                             | 183               | 2                   | 0                       | $1 - 2/183 = 0.989$                |
| 0.0520                             | 181               | 1                   | 0                       | $0.989 \times (1 - 1/181) = 0.984$ |
| 0.1040                             | 180               | 2                   | 0                       | $0.984 \times (1 - 2/180) = 0.973$ |
| 0.1314                             | 178               | 1                   | 0                       | $0.973 \times (1 - 1/178) = 0.967$ |

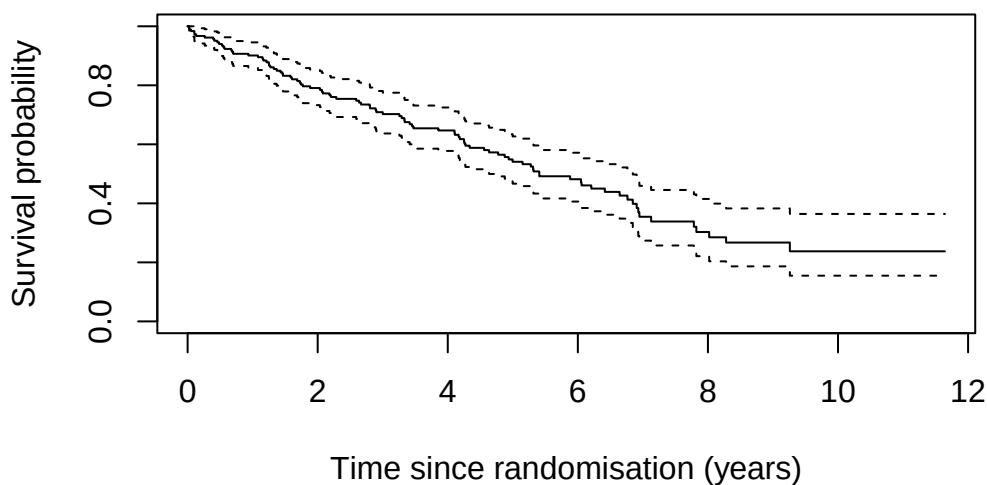
## Question 2

- Plot the Kaplan-Meier survival curve using `plot(pbc.km)` and look at how the curve is constructed from the table of times and probabilities. Add labels to the axes (using `xlab=" "` and `ylab=" "`) to make the plot more informative.
- Another way of creating the plot is using `ggsurvplot` in the `survminer` package. Take a look at what information this gives.

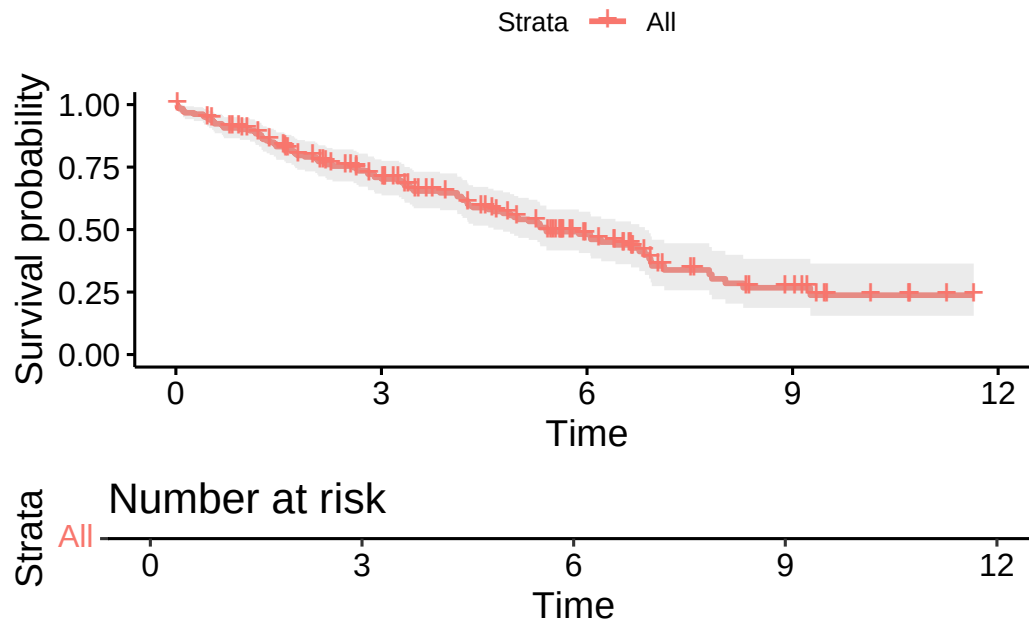
```
ggsurvplot(pbc.km, data = pbc, risk.table = T, conf.int = T)
```

## SOLUTION

```
#Plot Kaplan-Meier survival curve (with and without confidence intervals)
plot(pbc.km, conf.int=T,
     xlab="Time since randomisation (years)",
     ylab="Survival probability")
```



```
#another way of making the plot
ggsurvplot(pbc.km, data = pbc, risk.table = T, conf.int = T)
```



When using `ggsurvplot` we see check marks on the curve, which show where censorings occur (there is also an option to show these in the standard plot using `mark.time=T`). It also shows a risk table at the bottom, which summarises the number of people at risk over time.

### Question 3

Next we will describe the survival experience of the two treatment groups using the Kaplan-Meier method.

- Obtain a Kaplan-Meier plot of the estimated survival functions in the two treatment groups using:

```

pbc.km <- survfit(Surv(time,d)~treat,data=pbc)

plot(pbc.km,conf.int=T,col=c("black","grey"),
     xlab="Time since randomisation (years)", ylab="Survival probability")

legend ("topright",c("Placebo","Active treatment"),
      col=c("black","grey"))

```

- Summarise survival in the two treatment groups, and the difference in survival.

### SOLUTION

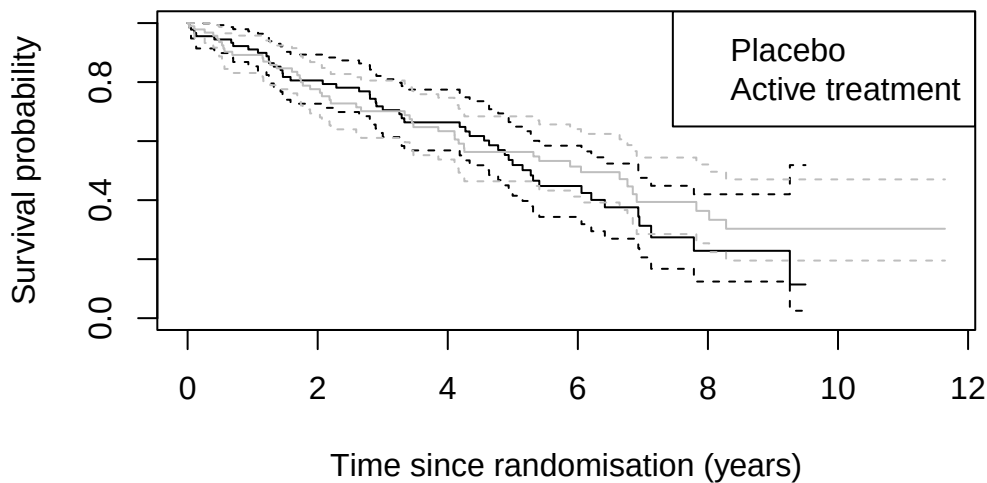
```

pbc.km <- survfit(Surv(time,d)~treat,data=pbc)

plot(pbc.km,conf.int=T,col=c("black","grey"),
     xlab="Time since randomisation (years)", ylab="Survival probability")

legend ("topright",c("Placebo","Active treatment"),
      col=c("black","grey"))

```



The estimated survival curves are similar in the two treatment groups up to about 2 years, then the survival curve in the active treatment group is slightly below that of the placebo group up to around 5 years, indicating that there was a slightly higher risk of death in the active treatment group during this period. From around 5 years onwards the survival curve for the active treatment group is above that for the placebo group, indicating lower risk of death in the active treatment group for those who survived up to about 5 years. However, the curves are generally very similar and the confidence intervals are largely overlapping over the whole time period. Note that the confidence intervals at the right hand part of the graph are wide because of the relatively few deaths which occurred later on. In both treatment groups around 50% of individuals survived to 5 years of follow-up.

#### Question 4

We will use the log rank test to formally compare survival curves in the two treatment groups.

- What is the null hypothesis?
- Perform the test using

```
survdif(Surv(time,d)~treat,data=pbcc)
```

- Interpret the results from the log rank test.

## SOLUTION

```
survdif(Surv(time,d)~treat,data=pbcc)
```

Call:

```
survdif(formula = Surv(time, d) ~ treat, data = pbcc)
```

|         | N  | Observed | Expected | (O-E) <sup>2</sup> /E | (O-E) <sup>2</sup> /V |
|---------|----|----------|----------|-----------------------|-----------------------|
| treat=1 | 90 | 49       | 45.3     | 0.297                 | 0.57                  |
| treat=2 | 94 | 47       | 50.7     | 0.265                 | 0.57                  |

Chisq= 0.6 on 1 degrees of freedom, p= 0.5

- The null hypothesis is that the survival curves are the same in the two treatment groups.
- The log rank test statistic is 0.57, which is compared with a chi-squared distribution with 1 degree of freedom, giving p-value 0.5. There is no evidence against the null hypothesis.

## Part B: Whitehall Study

We now return to the Whitehall dataset used in Practical 1. In this practical we will use the variables in the table below.

| Variable | Description                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id       | Unique identifier for each participant                                                                                                                            |
| timebth  | Date of birth                                                                                                                                                     |
| timein   | Date person entered the study                                                                                                                                     |
| timeout  | Date of the end of follow-up due to either death or censoring                                                                                                     |
| chd      | Event indicator at the end of follow-up: 0=alive or died from other cause, 1=death due to coronary heart disease                                                  |
| sbpgrp   | Systolic blood pressure (SBP) recorded at cohort entry, categorised as:<br>1: $SBP < 120$ ; 2: $120 \leq SBP < 140$ ; 3: $140 \leq SBP < 160$ ; 4: $SBP \geq 160$ |

Load the data using:

```
whl<-read.table("datasets/whitehall.csv",sep=" ",header=T)
```

## Question 1

First we will investigate survival in this cohort using time-in-study as the timescale, i.e. the time origin is the time of joining the cohort.

- Format the date variables as dates, like you did in Practical 1.
- How many deaths were there and what was the total person-time of follow-up, in years.
- Obtain a Kaplan-Meier plot and interpret the results. What is probability of survival beyond 10 years?

## SOLUTION

```
#format timein, timeout,and timebth as dates
```

```
whl$timein=as.Date(whl$timein,"%d%b%Y")
whl$timeout=as.Date(whl$timeout,"%d%b%Y")
whl$timebth=as.Date(whl$timebth,"%d%b%Y")
```

```
#number of deaths
table(whl$chd)
```

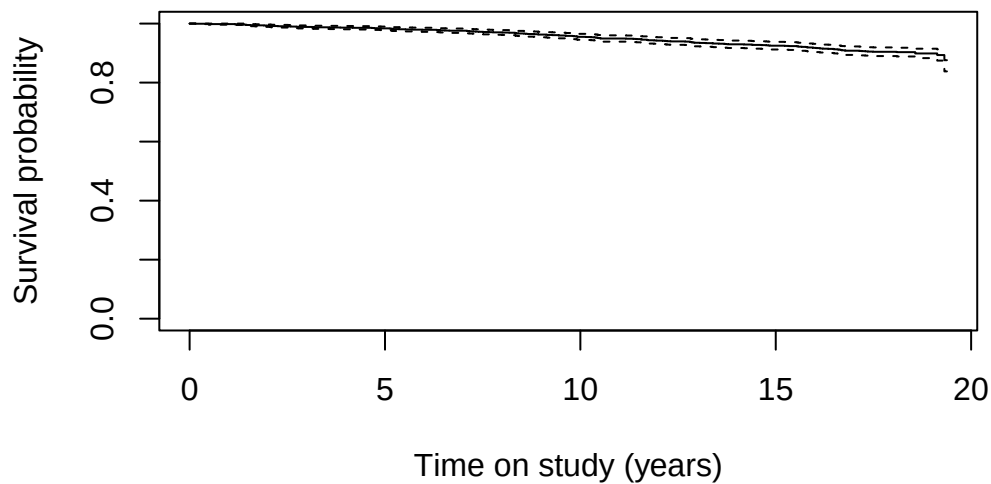
```
  0    1
1523 154
```

```
#total person-time of follow-up, in years
sum(as.numeric(whl$timeout-whl$timein)/365.25)
```

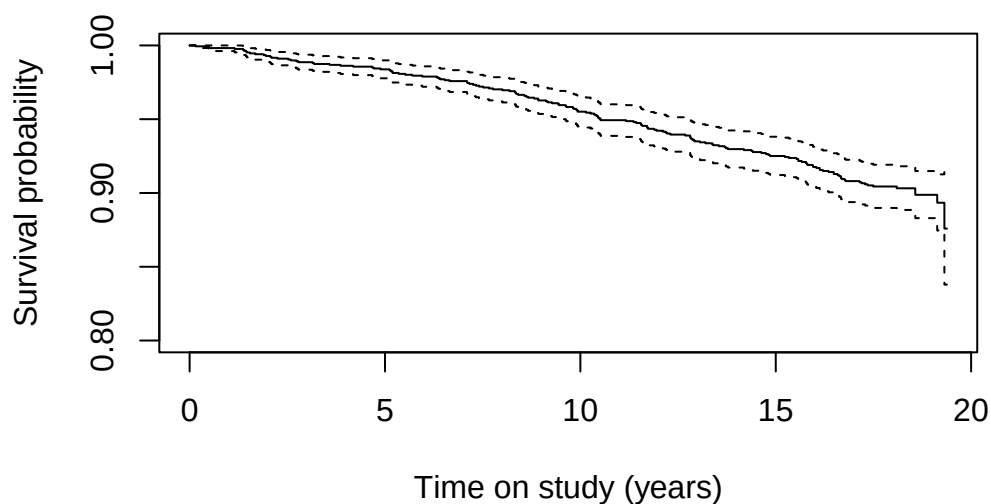
```
[1] 27603.3
```

```
#Kaplan-Meier analysis
whl.km <- survfit(Surv(time=(timeout-timein)/365.25,event=chd)~1,
                  data=whl)

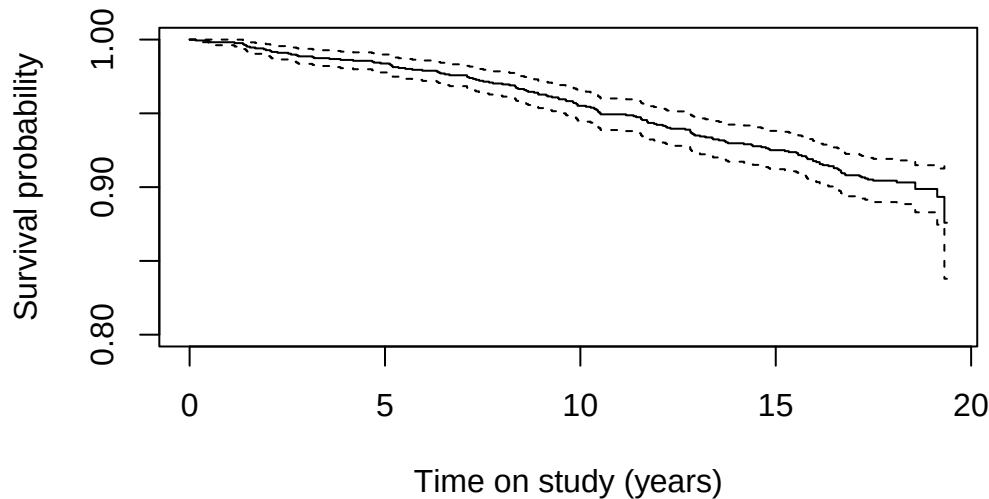
plot(whl.km,
      xlab="Time on study (years)", ylab="Survival probability")
```



```
#It can use useful to zoom in on the curve
plot(whl.km,ylim=c(0.8,1),
     xlab="Time on study (years)", ylab="Survival probability")
```



```
#An alternative way of specifying the timescale using Surv(),
# which gives identical results
whl.km2<- survfit(Surv(time=as.numeric(timein),
                      time2=as.numeric(timeout),
                      event=chd,origin=as.numeric(timein))~1,
                 data=whl)
plot(whl.km2,ylim=c(0.8,1),
     xlab="Time on study (years)", ylab="Survival probability",
     xscale=365.25)
```



```
#probability of survival beyond 5, 10, and 15 years
summary(whl.km, times=c(5, 10, 15))
```

```
Call: survfit(formula = Surv(time = (timeout - timein)/365.25, event = chd) ~
1, data = whl)
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 5    | 1614   | 27      | 0.984    | 0.00310 | 0.978        | 0.990        |
| 10   | 1497   | 46      | 0.955    | 0.00514 | 0.945        | 0.965        |
| 15   | 1379   | 46      | 0.925    | 0.00662 | 0.912        | 0.938        |

- There were 154 deaths and there was a total of 27603.3 years of person-time of follow-up.
- The Kaplan-Meier analysis estimates that the probability of survival beyond 10 years was 0.955 (95% CI 0.945 to 0.965).

## Question 2

We will now investigate how changing the time origin and the time period for which an individual is under observation (or ‘at risk’) changes the Kaplan-Meier plot. Obtain Kaplan-Meier curves for the following three different approaches and discuss what you see and how the plots can be interpreted:

- Time origin and start of period in which the participant is ‘at risk’: Date a participant entered the study
- Time origin and start of period in which the participant is ‘at risk’ = Participant’s date of birth

- (iii) Time origin = Date of birth; start of period in which the participant is 'at risk' = date a participant entered the study

## SOLUTION

```
#(i) Time origin and start of period in which the participant
# is 'at risk': Date a participant entered the study
whl.km1<-survfit(Surv(time=as.numeric(timeout)/365.25,
                    event=chd,
                    origin=as.numeric(timein)/365.25)~1,data=whl)

plot(whl.km1,
     xlab="Time in study (years)", ylab="Survival probability")

#(ii) Time origin and start of period in which the participant
# is 'at risk' = Participant's date of birth
whl.km2<-survfit(Surv(time=as.numeric(timeout)/365.25,
                    event=chd,
                    origin=as.numeric(timebth)/365.25)~1,data=whl)

plot(whl.km2,
     xlab="Age (years)", ylab="Survival probability")

# (iii) Time origin = Date of birth; start of period in which
# the participant is 'at risk' = date a participant entered the study
whl.km3<-survfit(Surv(time=as.numeric(timein)/365.25,
                    time2=as.numeric(timeout)/365.25,
                    event=chd,
                    origin=as.numeric(timebth)/365.25)~1,data=whl)

plot(whl.km3,
     xlab="Age (years)", ylab="Survival probability")

#youngest age at joining the cohort is 40,
# so these data only provide information about survival from age 40
# when we use age as the time scale
range(whl$agein)

plot(whl.km3,xlim=c(40,85),
     xlab="Age (years)", ylab="Survival probability")
```

- Approach (i) uses time-in-study as the timescale. Approaches (ii) and (iii) use age as the time scale, but (ii) does not account for the delayed entry/left truncation, which is

incorrect. In (ii) individuals are being considered to be ‘at risk’ before they joined the cohort, which we should not allow. If there is delayed entry we should account for that in the analysis.

- In the plot from approach (iii), which accounts for the left truncation, the survival probably is at 1 up until around age 40, because no one was observed until that age in the study, and hence no deaths were observed before that age. These data can provide information on mortality only for the age range over which people were observed. We could estimate from these data the probability of survival beyond age  $X$  conditional on living to age 40 (for  $X > 40$ ), even though not everyone was observed from age 40, provided we account properly for the left-truncation in the analysis.
- The time-in-study timescale would be appropriate if we aimed to predict subsequent risk of CHD using characteristics recorded at the time of entering the cohort. We could account for age at entry in models fitted on the time-in-study timescale. In some circumstances we might be interested in the age timescale, for example if we are interested in how individual characteristics are associated with the hazard for CHD at a particular age.

### Question 3

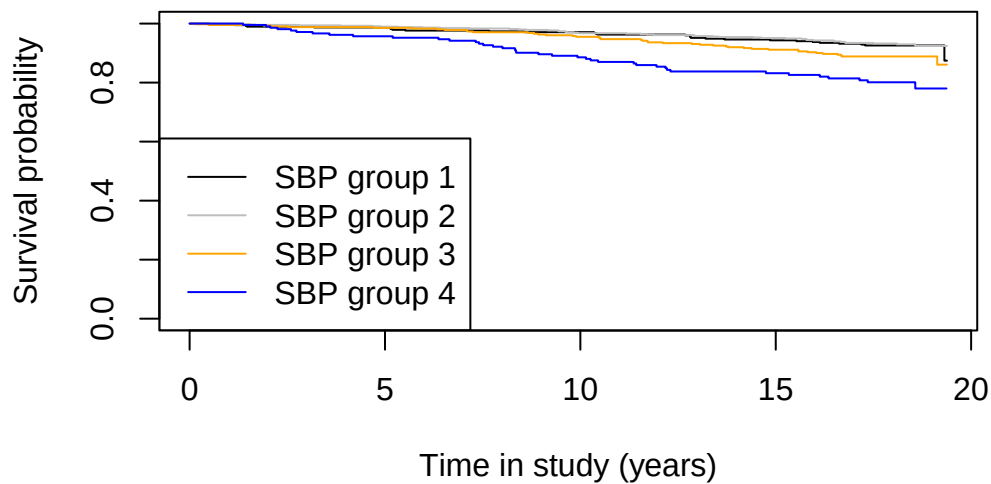
In this questions we will compare the survival of civil servants who had different levels of systolic blood pressure (SBP) at entry into the study, using the categorical variable `sbpgrp`.

- Obtain Kaplan-Meier estimates of survival in the four SBP groups, and compare.
- Use a log rank test to compare survival across the blood pressure groups. What is the null hypothesis? What are the degrees of freedom for the test? Interpret the results.

### SOLUTION

```
#Kaplan-Meier analysis
whl.km.sbp <- survfit(Surv(time=(timeout-timein)/365.25,event=chd)~sbpgrp,
                     data=whl)

#Plot
plot(whl.km.sbp,
     col=c("black","grey","orange","blue"),
     xlab="Time in study (years)", ylab="Survival probability")
legend (x="bottomleft",
       c("SBP group 1","SBP group 2","SBP group 3","SBP group 4"),
       col=c("black","grey","orange","blue"),lty=1)
```



```
#Estimates at 10 years
summary(whl.km.sbp, times=10)
```

```
Call: survfit(formula = Surv(time = (timeout - timein)/365.25, event = chd) ~
      sbpgrp, data = whl)
```

| sbpgrp=1     |          |          |          |          |                      |
|--------------|----------|----------|----------|----------|----------------------|
|              | time     | n.risk   | n.event  | survival | std.err lower 95% CI |
|              | 1.00e+01 | 3.58e+02 | 1.10e+01 | 9.71e-01 | 8.65e-03 9.54e-01    |
| upper 95% CI |          |          |          |          | 9.88e-01             |

| sbpgrp=2     |        |         |         |          |                      |
|--------------|--------|---------|---------|----------|----------------------|
|              | time   | n.risk  | n.event | survival | std.err lower 95% CI |
|              | 10.000 | 606.000 | 21.000  | 0.967    | 0.007 0.954          |
| upper 95% CI |        |         |         |          | 0.981                |

| sbpgrp=3     |         |          |         |          |                      |
|--------------|---------|----------|---------|----------|----------------------|
|              | time    | n.risk   | n.event | survival | std.err lower 95% CI |
|              | 10.0000 | 363.0000 | 18.0000 | 0.9551   | 0.0103 0.9351        |
| upper 95% CI |         |          |         |          | 0.9756               |

| sbpgrp=4     |         |          |         |          |                      |
|--------------|---------|----------|---------|----------|----------------------|
|              | time    | n.risk   | n.event | survival | std.err lower 95% CI |
|              | 10.0000 | 170.0000 | 23.0000 | 0.8857   | 0.0225 0.8427        |
| upper 95% CI |         |          |         |          |                      |

0.9308

```
#log rank test  
survdif(Surv(time=(timeout-timein)/365.25,event=chd)~sbpgrp,data=whl)
```

Call:

```
survdif(formula = Surv(time = (timeout - timein)/365.25, event = chd) ~  
        sbpgrp, data = whl)
```

|          | N   | Observed | Expected | (O-E) <sup>2</sup> /E | (O-E) <sup>2</sup> /V |
|----------|-----|----------|----------|-----------------------|-----------------------|
| sbpgrp=1 | 383 | 28       | 36.8     | 2.110                 | 2.78                  |
| sbpgrp=2 | 664 | 44       | 62.9     | 5.678                 | 9.60                  |
| sbpgrp=3 | 417 | 43       | 37.1     | 0.939                 | 1.24                  |
| sbpgrp=4 | 213 | 39       | 17.2     | 27.663                | 31.17                 |

Chisq= 36.4 on 3 degrees of freedom, p= 6e-08

- The estimated survival curves for groups 1 and 2 are similar. Those in the highest blood pressure group have the lowest survival curve (i.e. the worst survival).
- The null hypothesis is that the survival curves are the same across the four blood pressure groups. The degrees of freedom for the Chi-squared distribution used in the log rank test is the number of groups minus 1, which is 3. The log rank test gives a very small p-value, indicating strong evidence against the null hypothesis that the survival curves are the same across the four blood pressure groups.